

CAPSTONE PROJECT

NYKAA BEAUTY & SKINCARE AUTOMATION FRAMEWORK

Capstone Project Proposal & Specification

Selected Website: Nykaa

Website Link: <https://www.nykaa.com>

Why Nykaa Was Selected?

Nykaa was selected for this capstone project because it is a real-world beauty and skincare e-commerce platform with multiple dynamic workflows suitable for automation testing.

The platform also provides sufficient functional coverage to design approximately 10–15 automated test cases for each module, resulting in a scalable end-to-end automation framework.

Project Description

This project focuses on automating the major functionalities of the Nykaa beauty and skincare platform using the Playwright automation framework. The framework will validate core workflows including product discovery, filtering, navigation, cart operations, wishlist handling, checkout validations, and responsive UI behavior.

Technology Stack

Category	Technologies
Automation Framework	Playwright
Runtime Environment	Node.js
IDE	Visual Studio Code
Version Control	Git & GitHub

Reporting	HTML Reports & Allure Reports
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Planned Modules

Each module will contain approximately 10–15 automated test cases covering functional, UI, and validation scenarios.

1. **Homepage Validation Module:** This module focuses on validating homepage banners, navigation menus, featured products, offers section, and page loading behavior.
2. **Product Search Module:** This module validates product search functionality including valid search, invalid search, search suggestions, and search result verification.
3. **Brand & Category Navigation Module:** This module validates brand navigation, skincare category traversal, menu navigation, and URL verification workflows.
4. **Filter & Sorting Module:** This module focuses on validating filters such as category, brand, skin type, price range, ratings, and sorting functionalities.
5. **Product Details Module:** This module validates product images, pricing information, ingredients, product descriptions, ratings, reviews, and availability.
6. **Cart Management Module:** This module focuses on add-to-cart functionality, remove item workflow, quantity updates, cart total validation, and cart persistence.
7. **Wishlist Management Module:** This module validates wishlist add/remove functionality, wishlist navigation, and wishlist product management workflows.
8. **Checkout Validation Module:** This module focuses on checkout form validations, invalid input handling, mandatory field validation, address verification, and order summary workflows.
9. **Footer & Links Module:** This module validates footer links, policy links, social media navigation, and external link verification.

10. **Responsive UI Validation Module:** This module focuses on responsive UI validation across different screen sizes, devices, and browser viewports.

Conclusion

This project aims to implement a scalable and maintainable Playwright automation framework for validating real-world e-commerce workflows on the Nykaa beauty and skincare platform. The project will help improve practical knowledge in UI automation, reporting, framework design, GitHub collaboration, and end-to-end testing practices.